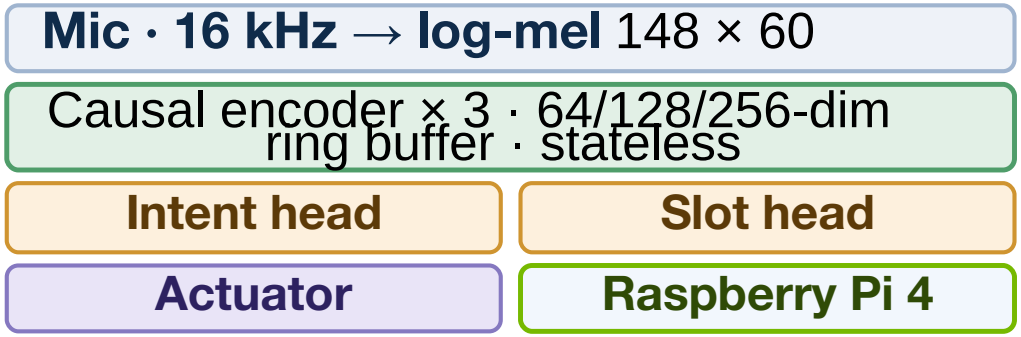
ME2 — Voice Controlled Smart Device

Always-on keyword + intent recognition at the edge · no cloud round-trip

1 October 2026 · Leandro Soneja · AI 231 ME2

Model



Item	Value
Parameters / weights	0.25 M · 0.52 MB

Validation on the Raspberry Pi 4

Item	Value
Keyword / intent acc	90.7% / 93.7%
False-accept rate	pending live test
Latency p95 / RTF	30.7 ms / 0.003
Runtime	TFLite f16 · 1 thread

Dataset

Item	Value
Source	own TTS corpus (V19)
Hours / utts	n/a h / 29,496
Speakers	21 TTS voices
Labels	19 intents · 3 slots

Training on the A100 cluster

Item	Value
Cluster	A100 ×8 shared, 1/run
Objective	CE + 0.1×CTC aux
Optimiser	Adam + cosine
Steps / loss	n/a / val 98.84%

To Be Submitted

GitHub repository	github.com/lcsoneja/me2-voice-controlled-smart-device
Dataset location	HF Drewboy/ai231-vcm-v20aux-phoneme-store (private, CC-BY-NC-4.0)
A100 cluster	8× A100-SXM4-40GB shared; seed 42
Model weights	GitHub Release v1.0-me2 + TFLite f16 + SHA256SUMS

What the reviewer will look for

#	Item	Status
1	Repo public, one-command reproduction	done
2	Dataset licensed and citable (DOI)	done
3	Training logs + final checkpoint committed	done
4	Pi 4 latency reproduced by the posted script	pending
5	Held-out test set, unseen speakers	done
6	Baseline of comparable size compared	done